Analysis and Investigation of Malicious DNS Queries Using CIRA-CIC-DoHBrw-2020 Dataset

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ABSTRACT Domain Name System (DNS) is one of the earliest vulnerable network protocols with various security gaps that have been exploited repeatedly over the last decades. DNS abuse is one of the most challenging threats for cybersecurity specialists. However, providing secure DNS is still a big challenging mission as attackers use complicated methodologies to inject malicious code in DNS inquiries. Many researchers have explored different machine learning (ML) techniques to encounter this challenge. However, there are still several challenges and barriers to utilizing ML. This paper introduces a systematic approach for identifying malicious and encrypted DNS queries by examining the network traffic and deriving statistical characteristics. Afterward, implementing several ML methods: (RF: Random Forest, DT: Decision Tree Classifier, GNB Gaussian Naive Bayes, KNN: k-nearest neighbor, Logistic regression, SVC: Support Vector Classifier, QDA: Quadratic Discriminant Analysis, SGD)". These models were employed to evaluate their ability to detect malicious DNS traffic using the CIRA-CIC-DoHBrw2020 data set. The Experiments revealed a good accuracy score where DT and RF models have achieved the highest accuracy, 99.99 % relative to other detection methods.

I. INTRODUCTION

DNS is an important protocol that has a substantial role in relation to web activities such as browsing and e-mail. It represents the phonebook of the Internet. Humans access information online through domain names, like google.com or yahoo.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources. DNS allows applications to use site names like Google.com instead of the IP addresses that cannot be memorized (Aijaz, Misbahuddin, & Raziuddin, 2020).

Since DNS is not used for Data transfer, numerous organizations get less consideration and have no monitoring plans in terms of security checking compared to other protocols like Web activity where attacks often take place. Since DNS is used by everyone, everywhere and all traffic flows through it. Point your traffic to the right destination. Because of its rule and its sensitivity in your real environment, it is exposed to many threats from attackers that target to control the DNS and grant them the possibility to abuse the DNS in order to extract and infect all data from it.

Since many organizations utilize one or two DNS servers, they may wake up to the reality that they are unable to protect their DNS against massive attacks with a large amount of traffic to their website that leads to servers crashing, preventing their users from accessing the website. This is because a large amount of traffic could be vulnerable to DNS security breaches. A malicious attack can also aim to exploit security vulnerabilities on the server that runs the DNS services and extract valuable data such as passwords, usernames, and other personal information.

Previous efforts in securing the DNS have focused on protecting the validity of information coming from the DNS. Since much of the Internet's traffic is encrypted and served by large content delivery networks, in many cases, domain name systems are the only clear text sign about the specific service being accessed.

DNS Tunneling is a method of cyber-attack that encodes the data of other programs or protocols in DNS queries and responses. DNS tunneling often includes data payloads that can be added to an attacked DNS server and used to control a remote server and applications.

Cyber criminals use several tunneling techniques to hide their identity. The most used techniques are FTP-DNS tunneling, HTTP-DNS, tunneling, HTTPS-DNS tunneling, and POP3-DNS tunneling.

Many traditional methods proposed to detect a malicious domain include the usual domain blacklist Names (Sammour, Hussin, & Othman, 2017), Network Traffic Analysis (Aiello, Mongelli, & Papaleo, 2013), Detailing of Web Page Content (Trejo et al., 2019), DNS Traffic Analysis (Zhao et al., 2019), and Analysis of salient lexical features (Allard et al., 2011). Most of the work on malicious URLs is content-based or non-content-based; the disclosure does not take into account the domain name and DNS data for the malicious account file URL, so the results obtained lack accuracy. Hence an effective mechanism for detecting the harmful field will also help in improving the accuracy of malicious URL detection.

Nowadays, several facilities are available for tunneling over DNS, and most of these tools point to free Wi-Fi access to sites that require restricted access via HTTP (Nadler, Aminov, & Shabtai, 2019). However, serious threats may occur with access to free Wi-Fi. These threats can be represented as malicious activities that can be assimilated through the DNS tunnel. With a DNS tunnel, complete remote control can be performed over a channel of a compromised internet host. Furthermore, various activities can be done via the DNS tunnel, such as file transfers system commands, or even IP tunnel. Feederbot (Ichise et al., 2018) and Moto (Almusawi, & Amintoosi, 2018) are examples of known DNS tunneling tools using DNS as a way to communicate.

All recently identified threats have stimulated the information security community to provide robust DNS tunnel detection methods (Farnham, & Atlasis, 2013). Different types of DNS tunnel detection techniques have been proposed. These methods can be classified into two broad categories; Traffic and payload analysis. The first category aims at analyzing the overall traffic as certain important features such as DNS traffic volume, number of hostnames per domain, site, and domain record can be identified. The second category analyzes the payload of a single query to identify many features such as content, number of bytes, and domain length.

The analysis of DNS tunnel features has led the researchers to use rule-based criteria where both traffic and payload are analyzed based on certain features. Once a pre-set condition occurs, DNS tunnel determination will be triggered. However, with the complex and time-consuming task of manual rules regulation, researchers tend to use machine learning techniques (MLT).

The main feature behind machine learning lies in the statistical model that has the ability to automatically define important rules (Nadler, Aminov, & Shabtai, 2019) (Almusawi, & Amintoosi, 2018). In addition, with the advent of annotated datasets such as JSON (Zang et al., 2020) containing network connections with predefined labels (such as Tunneled or Legitimate). The focus on machine learning has expanded because MLT requires annotated historical data. Hence, MLT is capable of training the model relying on this data. According to this training, new data will be available for testing.

In fact, there are many types of MLTs, like SVM, NB, DT, KNN, and others. With this diversity, it is difficult to determine which classifier would be more suitable, which would suit DNS tunnel discovery. This paper aims to [provide a comparative analysis of the DNS tunneling process using 9 MLTs classifiers, including NB, DT, and SVM.

This paper presents a lightweight approach leveraging ML models to detect malicious activities designed specifically to be deployed in the internal network of an enterprise. To detect malicious domains using a model trained by a machine learning algorithm using a combination of features of a domain name such as DNS data, lexical characteristics, and website reputation. We create separate data sets for benign and malicious domain names from various well-known and reliable sources and extract the above-mentioned features from those domain names and feed them to logistic regression machine learning algorithms and generate a model. We present an approach that demonstrates the simplicity, robustness, and scalability of our approach via empirical experiments on real-world data. The produced model is then experimented with a new list of domain names to classify them as benign or malicious.

The structure of this paper is as follows: section 2 includes a brief description of DNS, and the previous related works are in Section 3. Section 4discusses the methodology, the results are addressed in Section 5, and the conclusion of the work is in Section 6.

II. RELATED WORK

There has not been much research focusing on Malicious and encrypted DNS traffic. Current approaches are complex solutions and have inconsistencies during the processing phase (Das et al., 2017). This paper adopts a systematic approach for anomaly DNS queries that results in significant detection and less overhead in traffic processing.

Preston (2019) focused on the primary domain as a filter to classify the DNS traffic rather than the queries. The features have been extracted from subdomains from multiple groups. The author used supervised machine learning for examining DNS traffic and filter benign and malicious domains. However, this approach has a limitation of the inability to detect malicious queries in the main domain. In which the sub-domain is not enough for detecting the other types of attacks.

Das et al. (2017) presented a novel approach by focusing on semi-supervised learning for detecting DNS tunnels. Their technique learns the characteristics of normal DNS traffic and calculates the MSE between different sample classes to detect DNS tunnels. The authors focused on text query with a limited number of features to classify DNS queries using ML techniques such as k-means clustering to classify DNS concentrated on the all-TXT queries, and just used ten features whereas ML needs a lot of features for more learning to get

high accuracy, however, detecting all TXT queries is time-consuming.

While Palaniappan et al. (2020) used a logistic regression algorithm with lexical feature-based analysis to classify DNS queries to benign and malicious DNS domains, the authors only used four features and used the active DNS analysis as a filter. The proposed model achieved 60% accuracy. The main limitation of this model is focusing on a small dataset.

K. Shima et al. (2019) collected network traffic on a 1-day basis and only focused on the reflector traffic. After that, he extracted the features on the network level. they used SVM and to classify benign and malicious DNS servers. This mechanism suffers from an inability to handle encrypted communication.

C. Liu et al. (2019) focused on DNS tunneling and deep learning to detect malicious queries based on Byte-level. The model can extract all information in the entire DNS queries. However, they only focused on the sequential and structural data that was found in the initial request.

Banadaki et al. (2020) examined a new dataset called CIRA-CIC-DoHBrw-2020 using several ML algorithms such as (XG Boost, Gradient Boosting, and Light Gradient Boosting Machine). In addition, they investigate the important features. However, the preprocessing and optimization phase were unclear.

III. METHODOLOGY

The proposed method consists of four main phases; Data Collection, Feature Extraction, preprocessing, and model deployment. The Data Collection phase's purpose is to identify a benchmark dataset of DNS tunneling in order to facilitate the comparison among the classifiers. The Feature Extraction phase goal to exploit some features of payload and traffic analysis. The preprocessing phase purpose of validating the data as suitable input of the ML algorithm Figure 1 show the proposed Methodology. The Model deployment consists of training and testing phases by carrying out several ML algorithms such as SVM, NB, and DT.

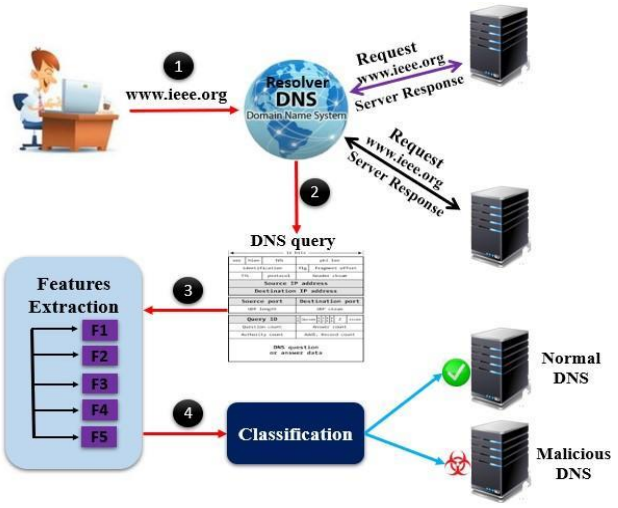

FIGURE 1. The proposed Methodology.

A. STEP 1: DATA COLLECTION

Finding an ideal dataset is a great challenge because it is considered private so it can be shared for privacy issues, does not reflect the current surface attack of cyber-attacks, and comes from different servers and operating systems which make it need normalization. In this paper, a realistic dataset was adopted from (Zang et al., 2020). It is new, encrypted data and follows a systematic approach in the generation phase. In addition, it contains malicious activities and has been compiled at the packet level, which helps in deep examination processes. The dataset includes raw data of malicious DNS traffic alongside normal DNS traffic.

The data collection process performed in four scenarios. The first scenario generating Non-DoH activity by accessing different web servers. The traffic has been collected using

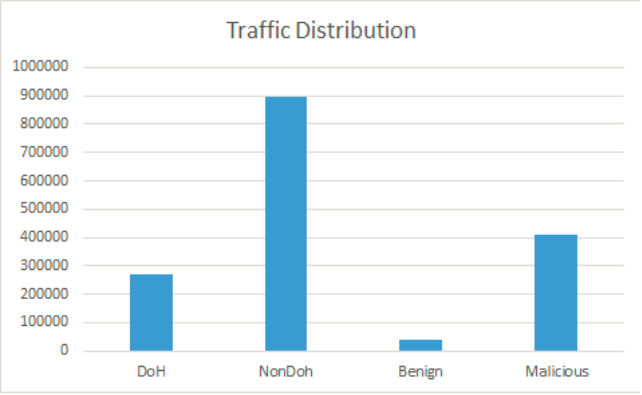

FIGURE 2. Traffic Distribution

wireshark and TCPdump to store it at the packet level, then convert the output to flow level in order to reduce the cost of processing resources. In the second scenario, Several DNS tunneling tools have been used such as DNSCat2, Iodine, and dns2tcp to generate Malicious-DoH traffic. These tools send TLS-encrypted HTTPS data in DNS queries to DoH servers (Adguard, Cloudflare, Google, Quad9). In the third scenario (Benign-DoH), Several web browsers have been used to generate Benign-DoH in the same mechanism as in scenario Non-DoH. In the fourth scenario, several browsers and DNS tunneling tools have been used to access the top 10k Alexa websites Figure 2 show the traffic distribution.

The public dataset available on the internet is usually unclean, incompatible, and sometimes suffers from several issues. Data preprocessing plays an important role in converting the unclean data into a clean and consistent format.

B. STEP 2: DATA CLEANING

This phase includes 3 main steps: removing the duplicate flows, handing the missing and outfitter values using the median, then encoding the categorical features (Source/Destination Port number/IP address) using one-hot encoding method.

C. STEP 3 FEATURE EXTRACTION

Two classes of features have been extracted from the adopted dataset using DoHMeter and CIC flow meter; statistical features such as mean, median average. and network features such as source IP, port number, flags. All extracted features are listed in Table I.

TABLE I
LIST OF EXTRACTED STATISTICAL TRAFFIC FEATURES

Parameter	Feature
F1-F4	Rate /Number of flow bytes sent or received
F5-12	Variance / Skew from median / Skew from mode /Coefficient of Variation/ Standard Deviation /Mean / Median of Packet Length
F13-F20	Standard Deviation/ Coefficient of Variation / Skew from median / Skew from mode /Variance /Mean/Median/ Mode of Packet Time
F21-F28	Median/ Standard Deviation / Coefficient of Variation / Skew from median/ Skew from mode /Variance /Mean/ Mode Request/response time difference
F29-F33	Source/Destination IP / Source/Destination Port number/ Timestamp
F34	Flow Duration

D. STEP4: DATA SCALING

This phase aims to normalize all features on the same scale to prevent biasing the ML models. Many types of scaling approach have been used in the literature. While data standardization method used in this paper.

E. STEP 5: DATA OVER SAMPLING

The adapted dataset is imbalanced as shown in the figure 2 which affects the reliability of the ML models. so, we used the SMOT algorithm for dataset oversampling.

F. STEP 6: DATA SPLITTING

The dataset has been split into training, testing, and validation. 80%, 10% and 10 % respectively. and use cross-validation with 10 folds.

G. STEP 7: MODEL DEPLOYMENT

This paper adapted eight ML methods as follow:

Random FOREST (RF): This algorithm is used in classification and regression problems. It is a type of a supervised classification algorithm and represents a collection of randomly selected DTs. The RF called ensemble learning and used to combine multiple classifiers in order to solve a complex problem and improve the performance of the classification model. RF collects the prediction accuracy of each tree and predicts the final output based on the majority votes of predictions. RF is considered a powerful algorithm since it has a lot of features that improve the results in the random forest such as runs efficiently on large databases, handles thousands of input variables without variable deletion, offers an experimental method for detecting variable interactions.

Decision Tree (DT): is a supervised learning technique and used in different fields such as statistics, data mining, and ML. it predicts responses values using learning decision rules derived from features. it can be used for decision making in both regression and classification tasks. DT consists of several components; the root node and branch node. or sub-tree, splitting, decision node, leaf or terminal node, pruning. DT is a powerful algorithm it is easy to understand, requires little data preparation, also able to handle numerical and categorical data, and deal with multiple output problems.

Logistic Regression (LR): is another technique of ML for regression algorithms. LR finds the relationships and dependencies between variables and transforms the output using the logistic sigmoid function to return probabilistic values that can be mapped to binary classes.

Quadratic Discriminant Analysis (QDA): is widely used in classification algorithms and statistics problems. It has a closed-form solution that can be easily computed with inherent multiclass, also has proven to work well in practice with no hyperparameters to tune. QDA is an extension of Linear Discriminant Analysis (LDA) which has a common variance for each class while in QDA, each class has its own variance or covariance matrix.

Support Vector Machines (SVM): is a supervised ML technique used for solving classification and regression problems. SVM generates a hyperplane in multidimensional space in an iterative manner to minimize the classification error rate. Moreover, SVM divides the datasets into classes to find a maximum marginal hyperplane. Accordingly, it achieves a high accuracy compared to other classifier models.

Stochastic Gradient Descent (SGD): this classifier basically is a simple and efficient optimization algorithm in ML and DL used to find the values of functions parameters that minimize the classification cost. Typically, there are three types of Gradient Descent; Batch Gradient Descent, Stochastic Gradient Descent, and Mini-batch Gradient Descent

Naive Bayes: A very simple and robust model of supervised ML that focuses on the application of Bayes' theorem with independent (naive) assumptions of conditionality between traits. Every feature is categorized independently of each other; hence it will speed up the prediction of the category of unknown data. Naive Bayes requires highly scalable features in a learning problem.

k-nearest neighbors: (KNN) algorithm is simple, easy-to-implement and not only used to solve classification problems but also regression problems. However, the flawed work for the KNN starts with selecting the number K of the neighbors then calculating the Euclidean distance of K number of neighbors and subsequently taking the K nearest neighbors as per the calculated Euclidean distance, and finally counting the number of the data points in each category and based on its distance between the input and the center of every class; then classifying the input into correct class (Al-Fawa'reh et al., 2020).

IV. EXPERIMENTS AND MODELS EVALUATION

The ML-based model is trained to detect the attack on network traffic as presented in Figure 1. The ML algorithm should be generic to detect the unseen instance correctly. Accuracy and ROC are generated using the number of correct predictions on the test dataset to find the actual class label against the predicted class label for each category and then extract the classification metrics. The accuracy represents the total correct prediction overall the total prediction, as shown in equation 1.

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

The ROC curve shows the relationship between the experimental sensitivity and the specificity for every possible cut-off. The ROC curve is a graph with the x-axis represents as shown in equations 2and 3:

$$1 - specificity\ (= false\text{-}positive\ fraction = FP/(FP+TN)) \quad (2)$$

While

$$y\text{-}axis\ represents\ TP/(TP+FN)) \quad (3)$$

The DT achieved the best results among all eight algorithms investigated for malicious and standard DNS queries with accuracy of 99.996% as shown in Figure 3 and Table II. However, RF and KNN does not deviate substantially from the highest accuracy. while LR, QDA, GNB and SGD have an average accuracy less than eight percentage points among the overall tested data sets. SVM is the worst algorithm in the training and testing time and SGD is the fastest algorithm in the testing phase.

The second part of the experiments evaluates these MLT's to discover their ability to identify the DNS traffic either encrypted or not. Figure 4 and Table III show the RF classifier has the best results among all other algorithms with 99.9802 % accuracy. The DT achieves the second-best accuracy with 999715%, while GNB and QDA are the worst algorithm with 91.5967% and 87.4713%, respectively. The conducted Experiments show that the SVM is the slowest algorithm in the training and testing phase, while GNB is the fastest algorithm in the training and the SGD is the fastest in the testing phase. To Summarize, the best-achieved results show a classification accuracy of 99.99% for RF, DT, KNN in M and B classification.

TABLE II
MALICIOUS AND BENIGN CALCIFICATION METRICS

Model	Training Time	Testing Time	Accuracy
RF	118.7859	0.586187	0.999802
DT	72.24614	0.098546	0.999715
QDA	5.168157	0.573868	0.915967
GNB	1.486871	0.513123	0.874713
SGD	4.446488	0.030216	0.934754
KNN	344.022	1536.963	0.99482
LR	28.24521	0.030461	0.936506
SVM	32743.5	40657.9	0.99917

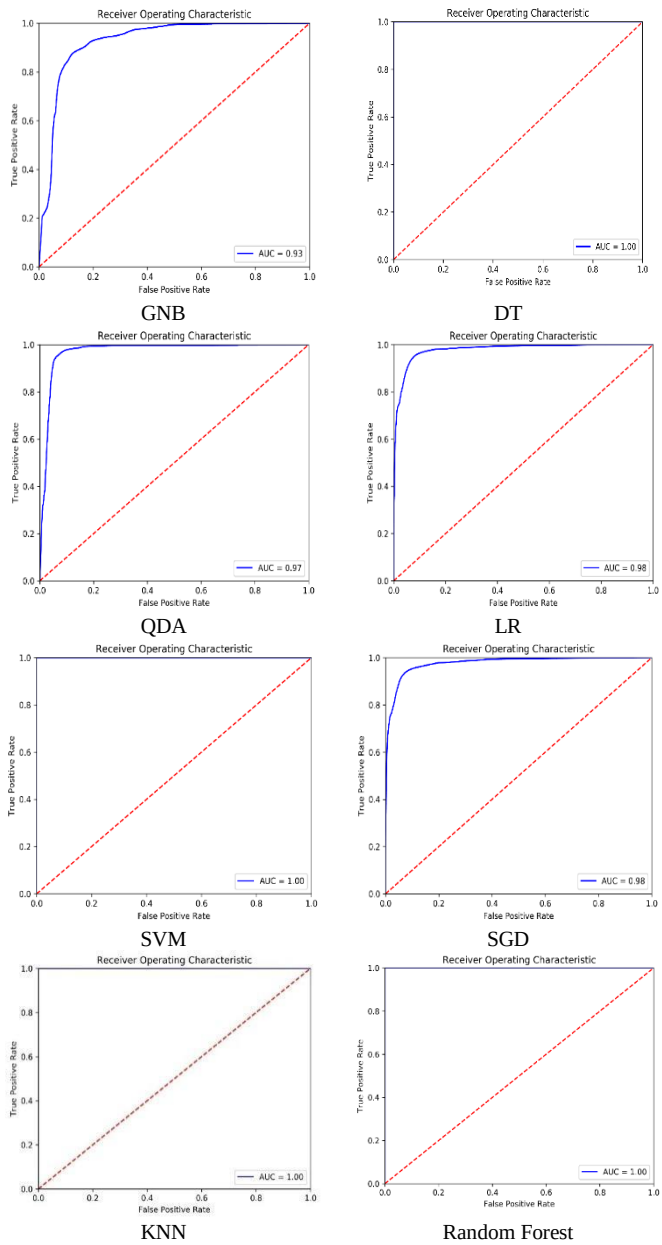

FIGURE 3. ROC Curve comparing MLT based on network and statistical feature for malicious and benign traffic

TABLE III
DOH AND NONDOH CLASSIFICATION METRICS

Model	Training Time	Testing Time	Accuracy
RF	31.69924	0.21619	0.999954
DT	11.90182	0.04168	0.99996
QDA	2.16018	0.26679	0.932362
GNB	0.70696	0.22751	0.799326
SGD	2.03135	0.01798	0.930879
KNN	95.52005	279.7582	0.999631
LR	11.40786	0.053378	0.935558
SVM	8185.87503	10164.49	0.99917

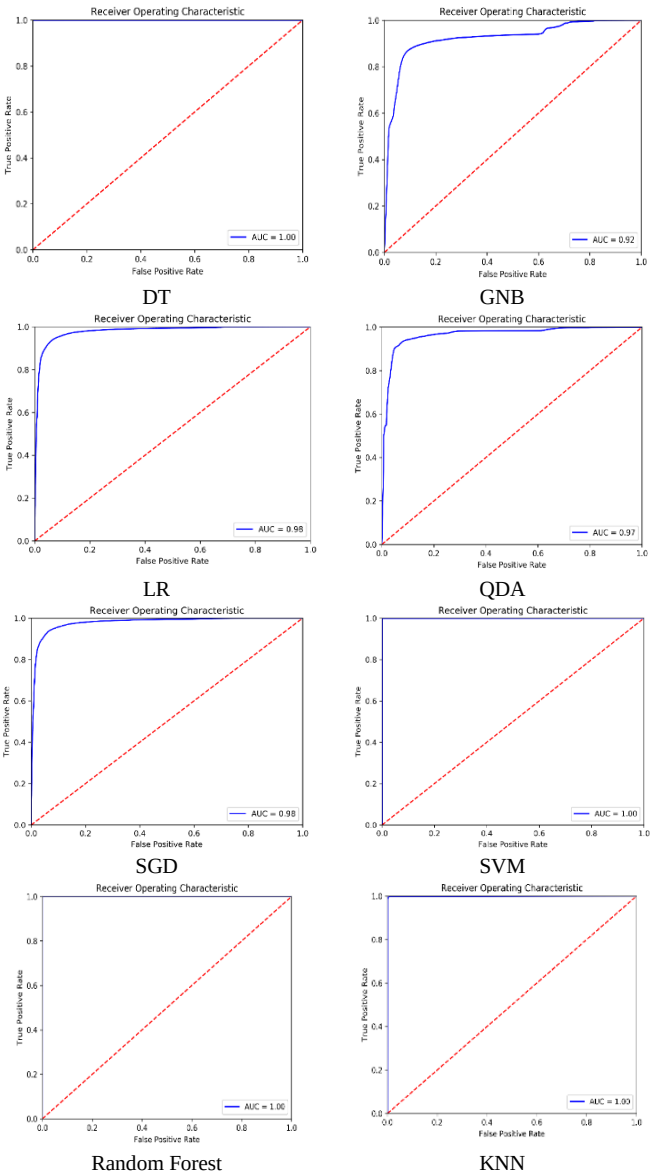

FIGURE 4. ROC Curve comparing MLT based on network and statistical feature for DoH and non-DoH traffic.

V. CONCLUSION

This paper has introduced a systematic approach for identifying malicious and encrypted DNS queries by examining network traffic using statistical characteristics. Several experiments have been conducted with eight ML algorithms to evaluate the efficiency and the performance of these classifiers using statistical and network features: All tests are experimented based on the CIRA-CIC-DoHBrw-2020 dataset. The attack rate of the dataset is approximately 25%. Other types of the collected traffic are distributed as 56% non-DoH traffic, 17% of Doh packets, and 2% of regular traffic. Experiments have shown that a single ML algorithm cannot effectively manage the time and the accuracy of detecting Malicious DNS queries. The accuracy of RF, SVM, DT, KNN is almost 99.9%. However, SVM and KNN are the slowest algorithms in the training phase. In contrast, GNB is the fastest algorithm for identifying traffic type but has the worst results in detection phase.

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